Zhiheng (Kevin) Li

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EDUCATION

University of Minnesota, Twin Cities Minneapolis, MN

*Bachelor of Science in **Mathematics** • **Computer Science Minor***

Expected May 2028

- **GPA:** 4.0/4.0 | **Honors:** Dean's List, Fall 2025, Spring 2026 · Tau Sigma Honor Society
- **Relevant Coursework:** Calculus I & II (A), Linear Algebra & Differential Equations (A), Physics for Scientists and Engineers I & II (A), Introduction to Programming Concepts (Python, A), Introduction to Statistical Analysis (A)

RESEARCH EXPERIENCE

SuperCDMS Dark Matter Experiment - School of Physics & Astronomy, University of Minnesota Minneapolis, MN

Undergraduate Research Assistant & UROP Fellow, Advisor: Prof. Yan Liu Oct 2025 - Present

• Built **cdms_spotlight**, a multilingual natural-language query system for the SuperCDMS DQM database hosted at **SLAC**: server-side filtering scaling from ~1,300 to ~100,000 series, cuts on any of 10,929 ODB parameters per series, self-learning local NLP (TF-IDF + logistic regression); runs 24/7 as a Slack bot in the official collaboration workspace, pip-installable CLI; presented at the SuperCDMS Software Meeting on July 14, 2026; contributed to the follow-up DQM/ODB upgrade, prompted by cdms_spotlight, that moves non-metadata parameter cuts server-side so queries no longer pull data to the client. Jul 2026

SNOLAB Run 4 Phonon Pulse Analysis May–Sep 2026

- Built the **phonon pulse template pipeline** for 13 SNOLAB R4 detectors (12 phonon channels each): event selection around the 10.37 keV activation line (~120 GB raw traces), free-pretrigger 2-exponential fits with quality cuts; the 1×1 and PCA N×M ROOT templates serve as the R4 pulse templates for optimal-filter reprocessing; presented at the SuperCDMS Analysis Working Group meeting on July 29, 2026. May–Jul 2026
- Measured the **phonon collection efficiency** of detector Z7: converted every K-line pulse from current to power with the TES electrothermal equations and integrated the fitted pulse in closed form, validated against the official absorbed-energy RQ; presented at the SuperCDMS Analysis Working Group meeting on September 23, 2026. Jul–Sep 2026

• Built **BlueFors-Alert**: minute-level PostgreSQL sync of the 64 GB CS2 database to a Raspberry Pi; Slack monitor with dual IDLE/COLD thresholds and time-range plots of any sensor; extended to a liquid-nitrogen scale. Jul 2026

• Extended RuOx (R31279 & R31839) **calibration curves** to 5 mK–300 K (PCHIP + Mott VRH); the generated .340 files are now in use on the lab's BlueFors dilution refrigerator. May 2026

• Built a **Python pipeline** for LED calibration data from 11 cryogenic Ge channels (5,200+ events, 10+ GB raw): baseline subtraction, SNR-based glitch detection, peak alignment, exponential fitting. Oct 2025

Mouse Eye Tracking for Neural Alignment - Deisseroth & Mitra Labs, Stanford University Remote

Research Collaborator (remote), with Kaiwen Sheng, Ph.D. student in Bioengineering Aug 2026 - Present

• Set up eye keypoint tracking with **DeepLabCut** for infrared videos of mice recorded over multiple days (eight points: pupil ellipse, eyelids, eye corners), and chose the training recipe on the first video by controlled experiments: batch size, backbone (ResNet-50) and the rule that picks frames for correction; wrote down the definition of every keypoint. Aug–Sep 2026

• Ran an **active-learning study on 19 videos from 3 mice**: each new video is pre-labeled by the previous model, corrected and retrained in steps of 20 labels until the error stops improving; for a new mouse, 20 labels cut the median error from 208 px to 10.4 px, most videos settled at 20–40 labels, and adding videos did not degrade earlier ones. Derived pupil size, eye opening and blinks frame by frame, unsmoothed, to align with brain recordings. Sep–Oct 2026

• Now: **finding the frames whose tracking cannot be trusted** without new labels, and fixing only those frames. Oct 2026

RESEARCH & INDUSTRY EXPERIENCE

Low-Carbon Transition & Industrial Monitoring Platform **Beijing / Shanghai, China**

Faculty-led Researcher (research phase) → Algorithm Engineer (industry phase), Shanghai Huaruizhongxin Technology Co., Ltd.

Oct 2024 - May 2025; Aug 2026 - Sep 2026

• Contributed to a faculty-led low-carbon transition project for China's steel industry, supporting a proposal and implementation plan for national ministry consideration through data collection, database construction, and quantitative analysis.

• During the industry implementation phase, contributed to an energy and carbon-emissions monitoring platform by designing a Python prototype for industrial time-series data quality governance, translating meter/point metadata and quality flags into rules for cumulative, instantaneous, consumption, enumeration, and statistical measurements.

• Implemented historical and real-time cleaning for range, constant, trend, missing-data, and communication anomalies; used same-time historical profiles for outage imputation and cumulative-to-consumption attribution; validated multi-day data with plots and regression tests, then packaged the work with Git version control.

SELECTED PROJECTS

College Students Entrepreneurial Project & Entrepreneurship Roadshow Beijing, China

Team Leader Oct 2024 - Jun 2025

- Led a student team to design **a billiards rest** that combines the high and low bridge heads into one;
- Took it from CAD model to 3D-printed prototypes, iterating through player testing and market research; received **national project approval with seed funding**; **Chinese utility patent** under CNIPA review.
- Invited to present at the **Tsinghua X-Lab & Kering Entrepreneurship Roadshow** to industry leaders.

AWARDS & HONORS

- **UROP** (Undergraduate Research Opportunities Program) Fellowship, University of Minnesota, Summer 2026
- **1st Prize**, Beijing University Student Innovation & Entrepreneurship Competition (Science & Technology Track)
- **1st Prize**, National College E-Commerce "Innovation, Creativity & Entrepreneurship" Challenge, Campus Level | Team Leader
- **3rd Prize**, China International College Students' Innovation Competition, Campus Level (2025) | Team Leader
- **2nd Prize**, NECCS National English Competition for College Students

ACTIVITIES & SERVICE

- **Special Olympics USA Games Minnesota 2026**, Tennis Division - Volunteer Ball Retriever | June 25, 2026
- **Chinese Student Union**, UMN - Active Member | Oct 2025 - Present
- **Chinese Students & Scholars Association**, UMN - Volunteer Calculus I Tutor | 2025 - Present
- **College Table Tennis Club / Football Club / Guitar Club** - Team Member | 2023 - 2024

PROFESSIONAL SKILLS

- **Techniques:** Cryogenic instrumentation, signal processing, data pipeline development, thermometer calibration curve fitting, real-time PostgreSQL sync, local NLP, Slack bot development, waveform template construction
- **Tools:** Python (NumPy, Pandas, SciPy, Matplotlib), scikit-learn, PostgreSQL, Java, ROOT, Git, Linux, Slurm, MSI HPC (Agate)
- **Languages:** Mandarin Chinese (Native), English (Fluent; IELTS 7.0)